Dr. Akhilesh Prasad, FRM, CQF

Quantitative Finance & Data Science

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Executive Summary

- An accomplished research-oriented professional holding a Doctorate in Quantitative Finance, an MBA in Finance, and a B.Tech. (Hons.) degree from IIT.
- Holds FRM and CQF certifications and has cleared CFA Level II and ERM Level I. Over 20 years of experience across corporate and academic sectors, with expertise in **Finance, Financial Engineering, and Data Science**.
- Coaching and mentoring researchers, students, and professionals online in **Quantitative Finance**, integrating the theoretical foundations of financial mathematics with hands-on applications of **data science and machine learning** in financial markets.

Key Strengths

Research	Quantitative Research, Financial Analytics, Volatility Modeling, Computational Finance
Technical	Python, Time Series Analysis, Machine Learning, Deep Learning, SQL, C/C++, Unix, and Data Structure & Algorithms
Finance	Financial Derivatives, Portfolio Optimization, Fixed-Income, Credit Derivatives, Risk Management, Probability & Stochastic Process, Stochastic Calculus, and Numerical Methods

Courses Taught

Finance	Fixed-Income, Corporate Finance, Portfolio Management, Derivatives
Quantitative Finance	Mathematics of Financial Derivatives, Quantitative Fixed-Income, Portfolio Optimization, Computational Methods in Finance, Stochastic Calculus for Finance
Data Science	Python, Time Series Analysis, Machine Learning, Deep Learning, AI for Business, Generative AI for Management, Mathematics

Educational Qualification

2024 – 2026 **Online, MSc in Financial Engineering**, *WorldQuant University*, USA
Thesis: A Fractional Option Pricing Model under Mandelbrotian Price Fluctuations

2019 – 2022 **Full-time, DBA in Quantitative Finance (accredited by TEQSA)**, *SP Jain School of Global Management*, Australia
Thesis: Forecasting Spikes in the CBOE VIX Index. (Link)

2014 – 2016 **Full-time, MBA in Finance**, *EDHEC Business School*, France
Thesis: Effect of Mergers and Acquisitions on Shareholders' Returns. (Link)

1997 – 2001 **Full-time, B.Tech (Hons.) in Agricultural & Food Engineering**, *IIT Kharagpur*, India
Thesis: Hydrologic Calculator, an Educational Interface. (Link)

Professional Qualifications

CQF	Certificate in Quantitative Finance, CQF Institute, Fitch Learning, UK
FRM	Financial Risk Manager Part I & II, GARP, USA
CFA	Chartered Financial Analyst Level I & II Passed, CFA Institute, USA
ERM	Enterprise Risk Manager Level I, Institute of Risk Management, UK

Academic Experience

03/2026 - **Associate Professor**, *Faculty of Management, SRM Institute of Science & Technology*, Chennai
Delivering courses in finance, analytics, and data science while performing several administrative duties.

02/2024 - 09/2025 **Assistant Professor**, *Jagdish Sheth School of Management*, Bangalore
Designed the Quantitative Finance discipline and taught several courses in Finance, Quantitative Finance, and Data Science.

11/2022- 01/2024 **Assistant Professor**, *IFMR GSB, Krea University*, Sri City
Developed and taught courses in quantitative finance.
Contributed to institution-building activities such as admissions interviews.

04/2022– 11/2022 **Assistant Professor**, *Jagdish Sheth School of Management*, Bangalore
Taught courses on Finance and Analytics and managed FinTech career track.

Visiting Faculty

06/2026 - **Invited Speaker**, *Department of Finance, Alliance University*, Bangalore
06/2026 Resource person for the international faculty development program on Machine Learning in Quantitative Finance

Industry Experience

2018 - 2019 **Finance Subject Expert**, *Study.com*, Remote
2016 - 2018 **Finance Subject Expert**, *Chegg India Pvt Ltd*, Remote
2010 - 2014 **Technical Team Leader**, *IBM India Pvt Ltd*, Pune
2008 - 2009 **Technical Lead**, *The Digital Group InfoTech Pvt Ltd*, Pune
2006 - 2008 **Senior Member Technical Staff**, *Oracle India Pvt Ltd*, Bangalore
2004 - 2006 **Senior Software Engineer**, *Patni Computer Systems Ltd*, Pune
2004 - 2004 **Contract Software Engineer**, *Eastern Software Systems Ltd*, New Delhi
2003 - 2004 **Assistant Software Engineer**, *Infocom Network Ltd*, New Delhi
2001 - 2002 **Consultant**, *SRIC & AgFE, IIT Kharagpur*, Kharagpur

Publications

2024 Shareholders' Wealth Dynamics in M&A Announcements in India. *Managerial Finance*. (Link) [**ABDC: B**]
2023 Forecasting Daily Changes in India VIX using DL. *IIMB Management Review*. (Link) [**ABDC: B**]
2022 Forecasting the Direction of Daily Changes in the India VIX Index Using ML. *JRFM*. (Link) [**ABDC: B**]
2022 Impact of U.S. Macroeconomic Variables on CBOE VIX. *JRFM*. (Link) [**ABDC: B**]
2022 Global Volatility Indices' Role in Predicting Indian Volatility. *Risks*. (Link) [**ABDC: B**]
2021 Machine Learning in Stock Market Investment Decisions. *Vikalpa*. (Link)
2019 Hydrologic Calculator for Hydrological Process Analysis. *CIGR Journal*. (Link)

Working Papers

Submitted Regime-Conditional Risk Factor Modeling of US Treasury Bond Portfolios: HMM, ICA, and Nonlinear Autoencoders
Submitted A Regime-Conditional ICA Framework for US Treasury Bond Portfolios: Stress-versus-Normal Factor Stability, VaR, and Hedging
Submitted A Semi-Implicit Caputo–Fabrizio Time-Fractional Black–Scholes PDE: Formulation and Empirical Calibration under Mandelbrotian Price Dynamics
Submitted Option Portfolio Construction Under Greek Constraints and Distributionally Robust CVaR: A Comparative Study
Submitted A Probabilistic PCA Framework for Term-Structure Risk Decomposition, VaR and Hedging: Comparative Evidence from Developed and Emerging Sovereign Bond Markets

Books

Writing Phase *Essential Portfolio Optimization with Python*
Writing Phase *Essential Mathematics for Finance, Economics, and Business*
Writing Phase *Essential Stochastic Calculus for Finance*
Writing Phase *Essential Artificial Intelligence for Business*: From Machine Learning to Generative AI

Technical Reviewer

2026 **Options Pricing with Python**, *Packt Publishing*
Bettaieb, M. *Options Pricing with Python: Master Option Pricing and Apply Python to Quantitative Finance, Trading, and Risk Management*. ISBN: 978-1-80730-199-6. [Amazon]

Patent

2002 Copyright for "Hydrologic Calculator". Reg. No. SW-858/2002

Accolades

2001 Proficiency Award for B.Tech. Thesis (AgFE), IIT Kharagpur
2000 First prize, Software Competition, Agricultural Engineering Society, IIT Kharagpur
2014 GMAT Score: 680 out of 800
1997 All India Rank 1860 in IIT-JEE

Personal Details

Date of Birth: March 01, 1977
Nationality: Indian
Marital Status: Married
Languages: English, Hindi
Address: A-702, IRIS, Magarpatta City, Hadapsar, Pune 411013, India